screamed downward, but not to the same extent as the volatility strategy, during the financial crisis.

How much should we allocate to volatility strategies? Given that there are pronounced (left-hand) tail risks, what sort of investor should pick up the small nickels and dimes in front a roaring steamroller?

2. Choices

Economists want agents (people, firms, or institutions) to have as many choices as possible and to guide agents to make optimal choices.[2] These choices reflect asset owners' preferences. Preferences are ultimately a collection of trade-offs: how do I balance a set of risks against a set of rewards? Trade-offs apply at a point in time, as in the choice between holding boring T-bills or picking up nickels in front of the oncoming steamroller in the volatility strategy. Trade-offs also occur across time, as in deciding how much to save today for retirement in the future. Preferences are unique to each asset owner, and ethics and morals (and sometimes lack thereof) can be important factors. Each asset owner has different goals, plans, and lifestyle choices. Psychological and family or institutional dynamics also play a role, and choices are influenced as well by the decisions of peers and other social factors.

We represent preferences by *utility,* an index numerically describing preferences in the sense that decisions that are made by ranking or maximizing utilities fully coincide with the asset owner's underlying preferences. (Economists call utility values "*utils*.") Utility is about choice, and we build a *utility function* to measure how satisfied an agent is by making choices.

Good choices in the context of optimal asset management policy start with the ancient Greek maxim "know thyself." Specifically, we build advice that depends upon how you feel during bad times. To capture this notion quantitatively, we define how the asset owner perceives risk and how she responds to it.

2.1. RISK

In his book, *Against the Gods: The Remarkable Story of Risk*, Peter Bernstein (1998) chronicles our astonishing progress in measuring risk through the development of statistics and probability from the invention of Arabic numerals to

[2] My colleague Sheena Iyengar (2010) at Columbia Business School shows that in some contexts, too much choice ends up being detrimental to consumers in making choices. I do not go down this route. Framing and other psychological contexts matter in adhering to optimal portfolio choice strategies as I discuss in section 2.6.